

BODY SYSTEMS

THE HUMAN BODY'S SYSTEMS WORK TOGETHER AS A TRUE COOPERATIVE. EACH SYSTEM FULFILLS ITS OWN VITAL FUNCTION, BUT ALL WORK TOGETHER TO MAINTAIN THE HEALTH AND EFFICIENCY OF THE BODY AS A WHOLE.

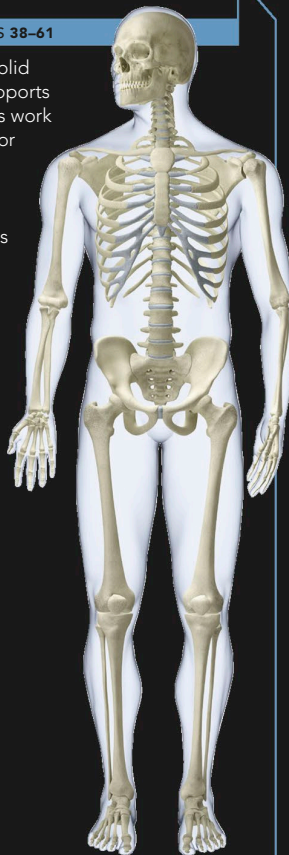
The exact number and extent of the body's systems is debated – the muscles, bones, and joints are sometimes combined as the musculoskeletal system, for instance. Although these systems can be described as separate entities, each depends on all of the others for physical and physiological

support. Most systems have some “general” body tissues, such as the connective tissues, which delineate, support, and cushion many organs. All the systems – except, and somewhat ironically, the reproductive system – are essential for our basic survival.

SKELETAL

EXPLORED ON PAGES 38–61

The skeleton is a solid framework that supports the body. Its bones work as levers and anchor plates to allow for movement. Bones also have a role in other body systems – blood cells develop in their fatty inner tissue (red marrow), for example. The body draws from mineral stores in bones during times of shortage, such as when calcium is needed for healthy nerve function.



MUSCULAR

EXPLORED ON PAGES 62–75

Muscles work in conjunction with the skeleton, providing the pulling force for varying degrees of movement, from powerful to finely tuned. Involuntary muscles work largely automatically to control internal processes, such as blood distribution and digestion. Muscles rely on nerves to control them and blood to keep them supplied with oxygen and energy.

